UWR Rainwater Offset Unit Standard

(UWR RoU Standard)

Concept & Design: Universal Water Registry

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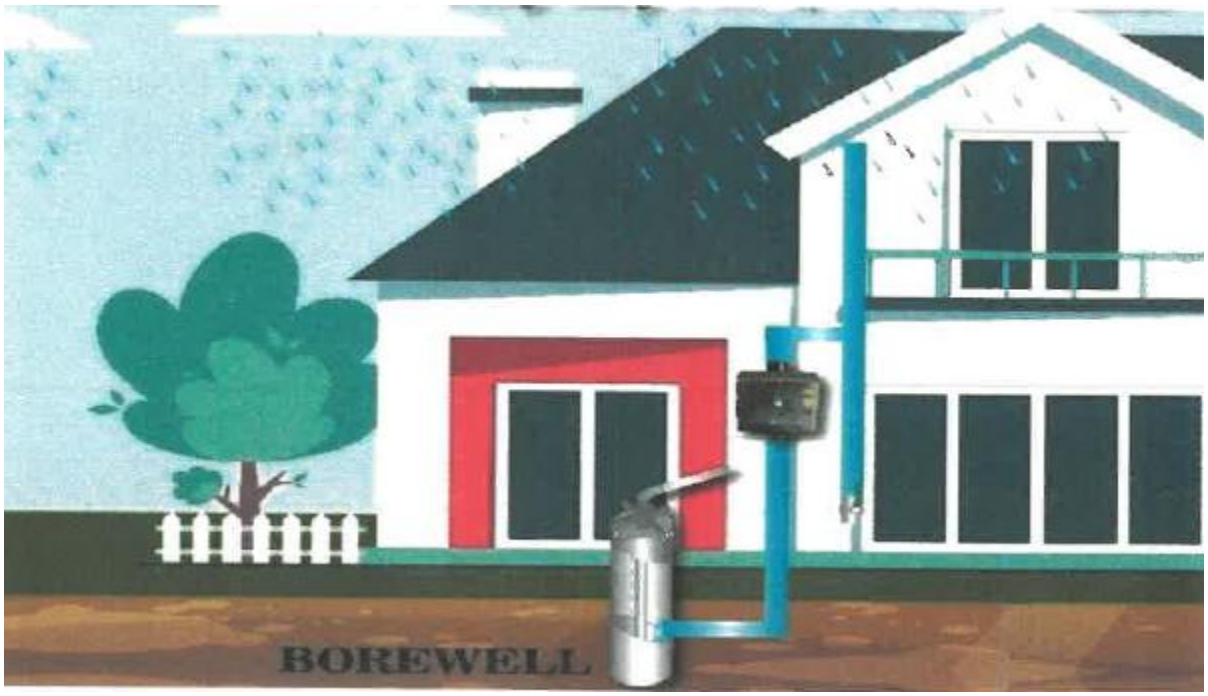

Project Concept Note & Monitoring Report (PCNMR)

Project Name: Borewell Recharge by The Surat-Tapi District Co-operative Milk Producers' Union Ltd., Gujarat

UWR RoU Scope: RoU Scope 2

Monitoring Period: 26/05/2025-31/12/2025

Crediting Period: 26/05/2025-31/12/2025

UNDP Human Development Indicator: 2 (India)

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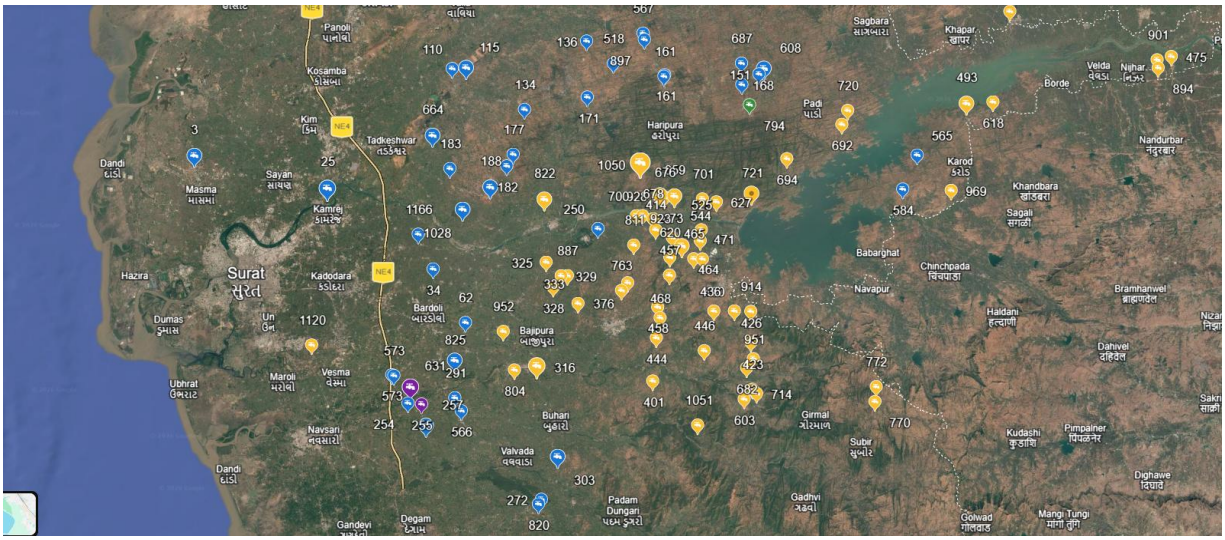
A.1 Location of Project Activity

Address of the Project Activity	**Surat-Tapi District Co-Operative Milk Producers' Union Ltd. (SUMUL)** Post Box No. 501, Sumul Dairy, Surat - 395 008, Gujarat, India
Project Proponent	The Surat District Co-operative Milk Producers' Union Ltd
State	Gujarat
District	Surat
Block Basin/Sub Basin/Watershed	Please refer to http://cgwb.gov.in/watershed/basinsindia.html Block basin: Tapi Sub Basin: Lower Tapi
Lat. & Longitude	21°12'18"N 72°50'24"E / 21.205°N 72.84°E
Primary water supply source	Borewell (Groundwater), Well and Storage Tank



A.2. Project owner information, key roles and responsibilities

The **Surat-Tapi District Co-operative Milk Producers' Union Ltd. (SUMUL)**, formerly known as The Surat District Co-operative Milk Producers' Union Ltd., is the **project proponent (PP)** and implementing agency for the rainwater harvesting and borewell recharge initiative across its member dairies. SUMUL is a district-level cooperative milk producers' union operating under the Gujarat Co-operative Milk Marketing Federation Ltd. (GCMMF), the marketers of Amul brand dairy products. In line with its long-standing objective of ensuring sustainable rural development and resource management, SUMUL has undertaken the installation of rainwater harvesting systems at its affiliated dairy units. At present, approximately 100 dairies under SUMUL have implemented rainwater harvesting infrastructure, wherein harvested rainwater is scientifically channeled for borewell recharge to enhance groundwater availability and long-term water security for dairy operations.

PP holds overall responsibility for planning, coordination, implementation, monitoring, and reporting of the project, while the individual member dairies provide the land, operational support, and maintenance of the rainwater harvesting structures within their premises. Clear lines of accountability are maintained, with **PP** acting as the nodal authority for technical guidance, compliance oversight, and consolidation of project information from all participating dairies.

The **PP**, hereby attests that it and its member dairies possess lawful water user rights for the areas within the project boundary and that the rainwater harvesting systems are used exclusively for groundwater recharge within the respective dairy premises. **PP** further confirms that the land on which the project infrastructure has been established is legally owned or lawfully held by the participating dairies and that there are no contested or disputed land titles within the project boundary. All necessary statutory permissions and approvals required for the implementation of rainwater harvesting and borewell recharge systems have been obtained or duly applied for in accordance with applicable local and state regulations. SUMUL also maintains complete and verifiable records of project implementation, including construction, installation, and associated operational expenditures.

Figure 1. 1. Water Harvesting Scheme

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The project has been implemented across **100 dairy society** locations distributed within Surat and Tapi districts of Gujarat, covering a wide range of talukas including Songadh, Mandavi, Umarpada, Mahuva, Valod, Bardoli, Vyara, Nizer, Kukarmunda, Kamrej, Mangrol, Olpad, Uchchal, and Dolvan. These locations represent both tribal and non-tribal regions, varying hydrogeological conditions, and differing levels of groundwater stress.

The selected sites correspond to functional Dairy Cooperative Societies (DCS) operating under The Surat-Tapi District Co-operative Milk Producers' Union Ltd. (SUMUL). Each site has been geo-referenced using latitude and longitude coordinates to ensure spatial traceability, facilitate monitoring, and support verification of the installed recharge structures.

The distribution of project locations demonstrates a decentralized implementation strategy, with installations spread across interior rural villages, semi-urban dairy clusters, and hilly tribal belts. This geographic spread allows the project to address groundwater recharge needs across diverse physiographic settings while ensuring equitable coverage within the operational jurisdiction of SUMUL.

Details of the individual project locations, including DCS code, village name, taluka, district, and geographic coordinates, are presented in Table below, which serves as the location register for the implemented recharge structures.

Sr. no.	DCS_Code	DCS_Name	Village	Taluka	District	Latitude	Longitude
1	473	Ukhalda	Ukhalda	Songadh	Tapi	21.2214815	73.4882673
2	464	Kelai	Kelai	Songadh	Tapi	21.2058317	73.502855
3	188	Kasal	Kasal	Mandavi	Surat	21.2913533	73.2081667
4	182	Uteva	Uteva	Mandavi	Surat	21.3258033	73.23355
5	115	Vakilpura	Vakilpura	Mangrol	Surat	21.4657404	73.1706658
6	171	Vahar	Vahar	Umarpada	Surat	21.4263952	73.3573059
7	687	Chavada	Chavada	Umarpada	Surat	21.4757962	73.5945649
8	608	Juna Chakara	Juna Chakara	Umarpada	Surat	21.4646831	73.6289496
9	794	Shaktipura	Shaktipura	Umarpada	Surat	21.4151875	73.6068467
10	664	Dharmpor	Dharmpor	Mandavi	Surat	21.3665806	73.1191851
11	952	Khambhala	Khambhala	Valod	Tapi	21.0842233	73.2286167
12	887	Vadi-Faliyu-Dhamodala	Vadi-Faliyu-Dhamodala	Valod	Tapi	21.165875	73.32709
13	1028	Jagruti-Mahila	Jaruti-Mahila	Mandavi	Surat	21.258695	73.1657149
14	254	Nihali	Nihali	Mahuva	Surat	20.9788	73.082035
15	257	Shri Gopala	Shri Gopala	Mahuva	Surat	20.9426483	73.11002
16	825	Dhundhesa	Dhundhesa	Mahuva	Surat	21.0381433	73.1541417
17	291	Kachal	Kachal	Mahuva	Surat	20.986865	73.1542417

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18	450	Jamankuva	Jamankuva	Songadh	Tapi	21.1144182	73.5520225
19	914	Khokhasa	Khokhasa	Songadh	Tapi	21.11387	73.6086633
20	446	Sandhkuva	Sandhkuva	Songadh	Tapi	21.0565255	73.5372596
21	303	Mahuvaria	Mahuvaria	Mahuva	Surat	20.8979183	73.3125283
22	3	Tadad	Tadad	Olpad	Surat	21.3360535	72.750816
23	682	Taparvad	Taparvad	Songadh	Tapi	21.0313554	73.6023337
24	820	Angaldhara	Angaldhara	Mahuva	Surat	20.83194	73.2829961
25	423	Mota Tarpada	Mota Tarpada	Songadh	Tapi	21.0001452	73.6111784
26	951	Temka	Temka	Songadh	Tapi	21.0449082	73.6126997
27	772	Virathava	Virathava	Songadh	Tapi	21.0040048	73.8007989
28	770	Malangdev	Malangdev	Songadh	Tapi	20.9823335	73.7987882
29	255	Dhorikuie	Dhorikuie	Mahuva	Surat	20.977385	73.1029783
30	493	Tapi-Khadakala	Tapi-Khadakala	Nizer	Tapi	21.4134681	73.9408683
31	714	Kapadbandh	Kapadbandh	Songadh	Tapi	20.9939707	73.617638
32	603	Vadi-Rupgadh	Vadi-Rupgadh	Songadh	Tapi	20.9857087	73.5985117
33	1166	SrijiDairy Farm	SrijiDairy Farm	Kamrej	Surat	21.225755	73.09733
34	136	Nasarapur	Nasarapur	Umarpada	Surat	21.5083012	73.3561435
35	804	Kosambia	Kosambia	Valod	Tapi	21.0279717	73.2459383
36	573	Tarbhon	Tarbhon	Bardoli	Surat	21.0190493	73.0603045
37	573	Tarbhon-Member	Tarbhon-Member	Bardoli	Surat	21.02014879	73.05709815
38	692	Amalpada	Amalpada	Songadh	Tapi	21.3865904	73.7488639
39	720	Mandavipani	Mandavipani	Songadh	Tapi	21.4069065	73.7578984
40	62	Motibhatlav	Motibhatlav	Bardoli	Surat	21.0969639	73.1703899
41	694	Satkashi	Satkashi	Songadh	Tapi	21.3366018	73.6642082
42	721	Juni Bavali	Juni Bavali	Songadh	Tapi	21.2827823	73.59677
43	161	Unchavan	Unchavan	Umarpada	Surat	21.4589039	73.4762835
44	161	Unchavan-Member	Unchavan-Member	Umarpada	Surat	21.456687	73.475308
45	1120	Kacholi	Kacholi	Choryasi	Tapi	21.0632633	72.933475
46	651	Akalutara	Akalutara	Kukarmunda	Tapi	21.551235	74.008175
47	618	Navibhilbhavali	Navibhilbhavali	Nizer	Tapi	21.4190039	73.9815635
48	659	Junai	Junai	Songadh	Tapi	21.2786267	73.491595
49	1050	Samarkuva	Samarkuva	Songadh	Tapi	21.320744,	73.439205

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50	923	Amaldi	Amaldi	Songadh	Tapi	21.2800617	73.4692133
51	676	Bhattwada	Bhattwada	Songadh	Tapi	21.2803433	73.476845
52	701	Shri Sarjamli	Shri Sarjamli	Songadh	Tapi	21.2780467	73.5344483
53	627	Limbi	Limbi	Songadh	Tapi	21.2737062	73.5559475
54	272	Kos	Kos	Mahuva	Surat	20.838805	73.28739
55	518	Chitalada	Chitalada	Umarpada	Surat	21.4754263	73.3975041
56	969	Sevati	Sevati	Uchchal	Tapi	21.2900713	73.9168109
57	414	Dhajamba	Dhajamba	Songadh	Tapi	21.2331111	73.4626759
58	928	Bhanpur	Bhanpur	Songadh	Tapi	21.253311	73.4328018
59	328	Kalamkui	Kalamkui	Valod	Tapi	21.1487617	73.305305
60	333	Khusalpura	Khusalpura	Vyara	Tapi	21.1253667	73.3435167
61	457	Nishana	Nishana	Songadh	Tapi	21.1667733	73.4838867
62	471	Dumda	Dumda	Songadh	Tapi	21.1903267	73.5343567
63	458	Chorvad	Chorvad	Songadh	Tapi	21.1200817	73.465805
64	620	Kavala	Kavala	Songadh	Tapi	21.192645	73.4844333
65	444	Dhamodi	Dhamodi	Songadh	Tapi	21.075125	73.4637083
66	525	Nani-Khervan	Nani-Khervan	Songadh	Tapi	21.232885	73.532715
67	544	Moti-khervan	Moti-khervan	Songadh	Tapi	21.2169367	73.53115
68	700	Panchpipla	Panchpipla	Songadh	Tapi	21.2535363	73.4425252
69	329	Dhamodala	Dhamodala	Valod	Tapi	21.1656517	73.317585
70	468	Kanara	Kanara	Songadh	Tapi	21.1044467	73.4694667
71	465	Amlipada	Amlipada	Songadh	Tapi	21.190695	73.52112
72	325	Bedkuva	Bedkuva	Valod	Tapi	21.18553	73.2943733
73	376	Katiskuva_Najik	Katiskuva_Najik	Vyara	Tapi	21.1556283	73.41982
74	401	Karnjvel	Karnjvel	Vyara	Tapi	21.0120783	73.4581767
75	678	Jamapur	Jamapur	Songadh	Tapi	21.2490214	73.458537
76	763	Bedkuva-Najik	Bedkuva-Najik	Songadh	Tapi	21.1443883	73.4102217
77	1051	Kelvan	Kelvan	Dolvan	Tapi	20.9479653	73.5271206
78	822	Kakadava	Kakadava	Dolvan	Tapi	21.2737122	73.2915871
79	436	Tokarava	Tokarava	Songadh	Tapi	21.1145883	73.58355
80	426	Gopalpura	Gopalpura	Songadh	Tapi	21.0686567	73.608935
81	34	Ishanpor	Ishanpor	Bardoli	Surat	21.1750381	73.1204435
82	316	Nalotha	Nalotha	Valod	Tapi	21.0287733	73.2803917

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83	177	Khodamba	Khodamba	Mandavi	Surat	21.3428417	73.2433851
84	183	Vadesia	Vadesia	Mandavi	Surat	21.3218182	73.1459381
85	250	Moticher	Moticher	Mandavi	Surat	21.2345233	73.373995
86	566	Karchelia	Karchelia	Mahuva	Surat	20.9680000	73.1642617
87	631	Kavitha	Kavitha	Mahuva	Surat	20.99647	73.085975
88	565	Gavan	Gavan	Uchchal	Surat	21.3416955	73.8646076
89	897	Kalijaman	Kalijaman	Umarpada	Surat	21.5104843	73.445107
90	567	Servan-Fokadi	Servan-Fokadi	Umarpada	Surat	21.5195735	73.4431534
91	25	Amboli	Amboli	Kamrej	Surat	21.28673	72.957033
92	168	Sevalan	Sevalan	Umarpada	Surat	21.4446617	73.595065
93	584	Amkuti	Amkuti	Uchchal	Surat	21.292026	73.8422812
94	901	Gaybeeumar	Gaybeeumar	Nizer	Tapi	21.479995	74.2355217
95	475	Devara	Devara	Nizer	Tapi	21.48493	74.2577083
96	894	Ranikhdakala	Ranikhdakala	Nizer	Tapi	21.468335	74.23708
97	151	Navachakra	Navachakra	Umarpada	Surat	21.4600493	73.6213547
98	811	Chikhali-Bhensrot	Chikhali-Bhensrot	Songadh	Tapi	21.2106959	73.4289973
99	110	Mangrol	Mangrol	Mangrol	Surat	21.4678425	73.1491156
100	134	Ognisa	Ognisa	Mangrol	Surat	21.4083216	73.2606117

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A.2.1 Project RoU Scope

PROJECT NAME	Borewell Recharge by The Surat-Tapi District Co-operative Milk Producers' Union Ltd., Gujarat
UWR Scope: RoU Scope 2	Measures for conservation and storage of unutilized water for future requirements including freshwater.
Date PCNMR Prepared	11/02/2026
Catchment Area	
Total catchment Capacity	
Type of structure	Rectangle shape
Month and Year of Construction	2025
Month and Year of Commissioning Rwh's	2025
Average Rainfall	**Rainfall Data** Tapi: 1968.57mm Surat: 1986.3mm
Run off Coefficient	0.9
RoU Crediting Period	26/05/2025 - 31/12/2025 (1 years)
Total RoUs Generated For the Crediting Period	**25953 RoUs**

A.3. Land use and Drainage Pattern

A3.1 Urban and rural Residential

A3.1.1 Introduction

Surat District is situated in the southern part of the state of Gujarat and extends approximately between latitudes 20°09' N to 21°23' N and longitudes 72°33' E to 73°30' E. The district covers a geographical area of about 4,549 square kilometres and forms part of the west-flowing Tapi river basin, ultimately draining into the Arabian Sea. As per the Census of India 2011, Surat District has a population of approximately 6.08 million, making it one of the most populous districts in the state.

Surat District is an administrative division in the southern part of the Indian state of Gujarat, with Surat serving as its headquarters. Known as the "Diamond City" and "Textile Hub of India," the district is a vital center for industry, trade, and coastal culture in western India.

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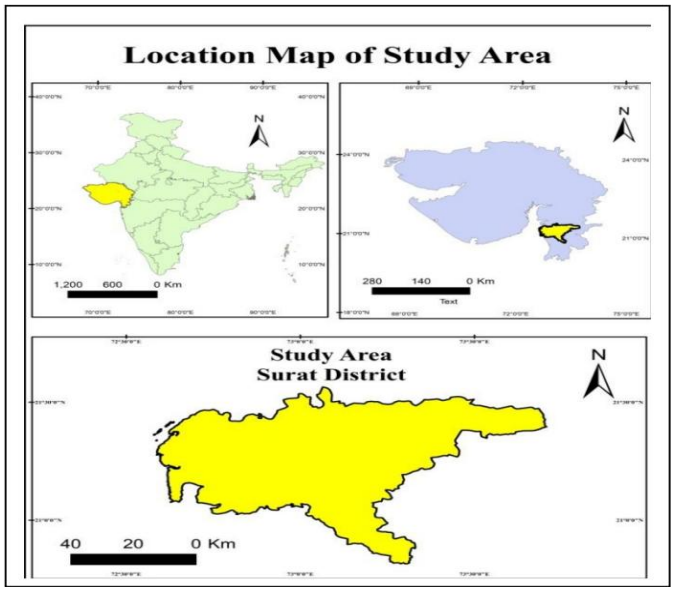

Figure 2. Location Map

The district comprises a mix of urban, semi-urban, and rural settlements, with Surat city functioning as a major urban and industrial centre, while the surrounding talukas support agriculture, allied activities, and cooperative-based dairy development. The district's physical setting, climatic conditions, river systems, and population distribution together influence land use patterns, drainage characteristics, and groundwater recharge potential.

Surat District is one of the fast industrializing regions of the State which has many industrial zones like Hazira in the west, Olpad in north-west and Surat city itself. These industries manufacture different types of chemicals, petrochemicals, polymers, fibres, etc. (GSIDS, 2016). The physico-chemical parameters study could help to determine the water quality. WQI calculation and Geographic Information System (GIS) has emerged as a powerful tool for storing, analysing and displaying spatial data and this data for decision making in several fields including environment, earth and engineering sciences, urban planning, agriculture, water resources etc. GIS is used as an effective tool for solution for water resources problem for assessing and mapping of groundwater quality.

https://www.isdesr.org/wp-content/uploads/2020/07/1.-Mukesh-Kumar-Bind-and-Rolee-Kanchan.pdf

A3.1.2 Physiography

Geographically, the district belongs to the western coastlands of the Deccan peninsula. The main Sahyadri scarp is a little outside the limits of the district towards east, but it gives the district its orientation, landscape features and drainage pattern. Distinct zones, viz., hilly areas, piedmont slopes, alluvial plains and coastal plains.

Hilly areas: The north-eastern parts of the district fall in this category. Here the general elevations are more than 100m amsl. The topography is rugged with low to moderate high hills and steep hill slopes. These parts are poorly populated and are infested by dense jungle of teak and bamboo.

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Piedmont slopes: East-central parts of the district fall in this category. Here the elevations range between 60 and 100m.amsl. These parts show a gentle slope towards the west. Topography is mainly plain with moderate to deep cutting river valleys and occasional hillocks.

Alluviual plains: Alluvial plains towards situated in the central parts of the district are characterized by flood plains of the Tapi, Kim and Purna rivers. Tapi has a meandering channel entrenched fairly deep and has cut deep terraces. The topography is generally plain with gentle slope towards west. The general elevations are below 60 m.amsl, the lowest elevation being 45m.amsl near Madhi.

Coastal Plains: The alluvial plains towards west merge into a dry barren sandy coastal plain fringed by marshy shore line. All the rivers form estuarine mouths. There are sand bars and spits near the shore.

https://www.cgwb.gov.in/old_website/District_Profile/Gujarat/Surat.pdf

A3.1.2 Geology

Major geological formations exposed in the district are Quaternary alluvium, Tertiary limestone and sandstones and Deccan Trap basalt. Figure -3 shows the geology of the district and the succession of geological formations in the district, as given by Geological survey of India in their brochure titled „" Know Your District-Surat"" is as follows.The geology of the district is depicted figure 3 ;

Geological Succession

Era	Period	Epoch /Series	Description
Cenozoic	Quaternary	Holocene	Soil, Younger alluvium Blown and Fluvio-marine deposits.
	Tertiary	Pleistocene	Older alluvium, conglomerate
		Miocene	Ferruginous sandstone (Gaj)
		Eocene	Numulitic limestone
		Palaeocene	Sub-Numulitic limestone, gypseous shale,Supra- Trappean sediments lateritised and having bauxite pockets.
Mesozoic	Lower Tertiary to Upper Cretaceous	Early Eocene To Upper	Deccan Trap basalt with dykes

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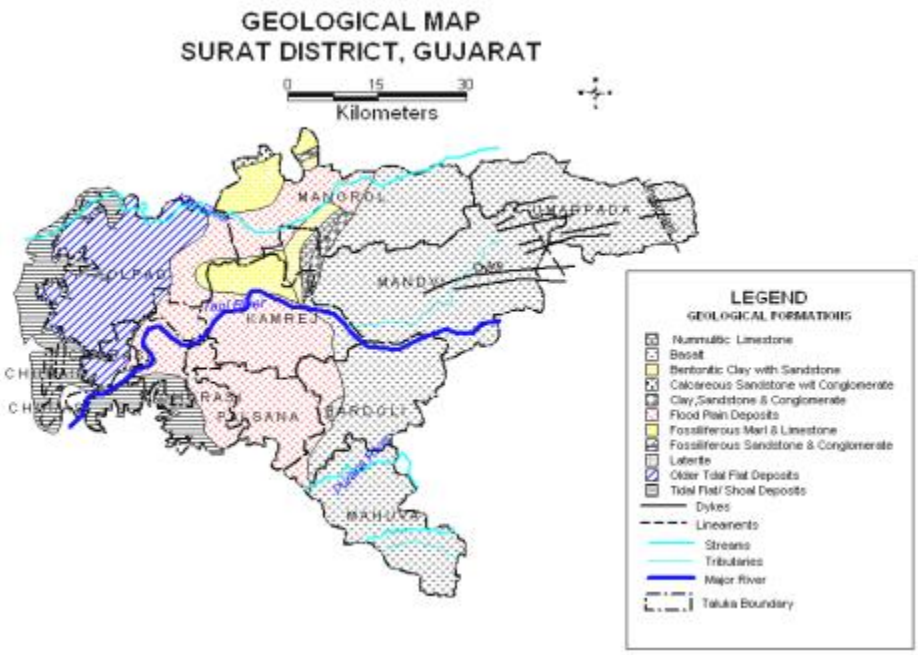

Figure 3. Geological Map Surat

https://www.isdesr.org/wp-content/uploads/2020/07/1.-Mukesh-Kumar-Bind-and-Rolee Kanchan.pdf

A3.1.3 Hydrogeology

Aquifer System

The hydrogeological framework of the area is essentially governed by geological setting, distribution of rainfall fall and facilities of circulation and movement of water through inter-connected primary and secondary porosity of the geological units forming the aquifers. The major aquifers in the district are formed by alluvium and Deccan Trap basalt with Tertiary formations occupying a small patch. The alluvium occurs in the western part of the district and along the streams whereas in eastern parts weathered and fractured basalt form aquifers. hydrogeological units: 1. Fissure Formations: 2. Porous Formation: Fissure Formations Deccan Traps from the aquifers in north-eastern, eastern and south eastern parts of the district comprising Mangrol, Mandvi, Vyara, Valod Mahuva, Songadh, Uchchhal and Nizar talukas. The ground water occurs in unconfined to semi-confined conditions. The occurrence and movement of ground water is governed by the thickness and extent of the weathered zone and presence of fractures and joints. At places, dykes act as ground water barriers and restrict the flow of ground water. On the other hand, at places, the dykes themselves form good aquifers owing to deeper levels of weathering in them. Depths of dug wells in Deccan traps range from less than 2m to 25m with maximum number of wells, about 80%, being 5 to 15m deep. The depths to water levels in the dug wells range from 1 to 20m. However, in 90% of the wells, the water levels are less than 10m. bgl. Yields of the dug wells are low to moderate in central parts whereas they are low to very low on the hill slopes. The dug wells generally sustain pumping for 4 to 5 hours at the yields of 50 to 150 m3 /day. Recuperation of water levels is generally slow. The western and north western parts of the district comprising Choryasi, Olpad, Kamrej, Palsana and parts of Bardoli talukas are covered by Alluvium. This aquifer can broadly be demarcated into two zones namely newer

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alluvium and older alluvium. The newer alluvium is present along the river courses and comprises fine to coarse grained sand trap wash with cay intercalations. The sand is unconsolidated but shows some degree of cohesion at places. Water levels are in general deeper in newer alluvium. Older alluvium is present in inter river plains and comprises sand, clay, kankar, grave and silt. The ground water occurs mostly under unconfined conditions but at places semi-confined conditions are also observed, probably due to the presence of clay lenses. The depths of the wells in alluvium generally range from 3.0 to 30.0m with some of the wells having bores down to 50m below bottom of the wells. The maximum number of wells, about 80%, are 5 to 20m deep. Depths to water levels in the wells range from 0.5 to 15m bgl about 90% wells show water levels less than 10m bgl. Yields of the dug wells and dugcum- bored wells range between 100 and 450 m^3/day.

Aquifer Parameters

The aquifer parameters data available with CGWB from the exploration carried out in different parts of the district was taken into the consideration. 8 In Alluvial areas the depth of the tube wells range from 36 to 169m with piezometric levels varying between 2 to 10m. The specific capacity of the wells ranges from 19 to 1658 m 3 /day/m. The transmissivity values obtained at these tube wells ranges from 104 (Kosad) to 2965m2 /day (Vihan).

Behaviour of Water Levels

The behaviour of water levels was studied based on the water level data collected from the National Network of Hydrograph Stations (NNHS). There are a total number of 23 monitoring stations in the district. Their data has been used for preparation of various hydrogeological maps. The water level data for 2012 of NHS established in the district is given in Annexure-II.

(A) Depth to Water Level

The map depicting the depth to water level (Figure 4 & 5) has been prepared based on water level data for May and November 2012 collected from NNHS.

The depth to water level ranges from 2.48m (Palsana) to 17.75m (Mahuva 2) during May2012. Spatially, the depth to water level in the district ranges between 2m to 20m during May 2012. The major area of the district is covered by the water level of 5m to 10m bgl . There are seven isolated pockets in the district which shows the water level ranging between 2 to 5m. Similarly, there are four isolated pockets in the district which show the 9 water levels ranging between 10 to 20m. There is no area where the water level of less than 2m and more than 20 meters is seen (Figure -4).

The depth to water level ranges from 0.1.39m (Palsana) to 16.55m (Mahuva2) during November2012. Spatially, the depth to water level in the district ranges between less than 2m to nearly 20m during November 2012. In the major part of the district the water level ranges between 2 m and 5 m bgl. The water level ranging between 5m and 10 m bgl is found considerably in a larger part of the district. There are three isolated pockets in the district where the water level of less than 2 m is also observed. Similarly water level ranging between 10 to 20m is seen in three isolated pockets in the district (Figure-5).

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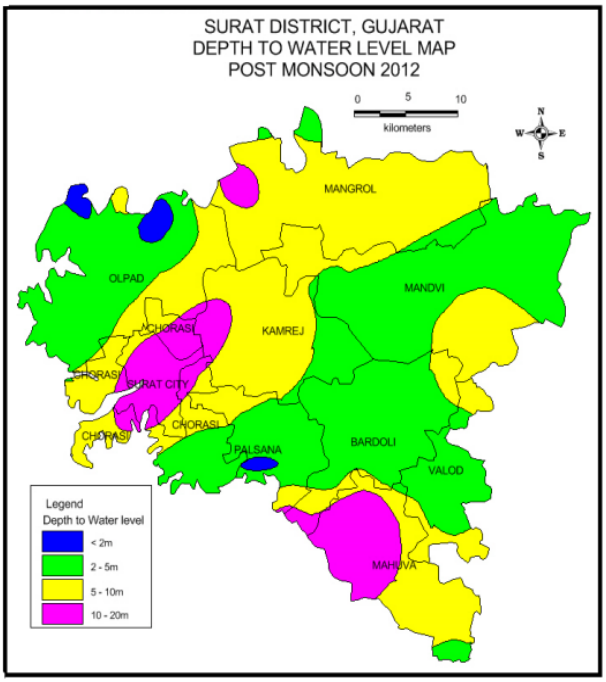

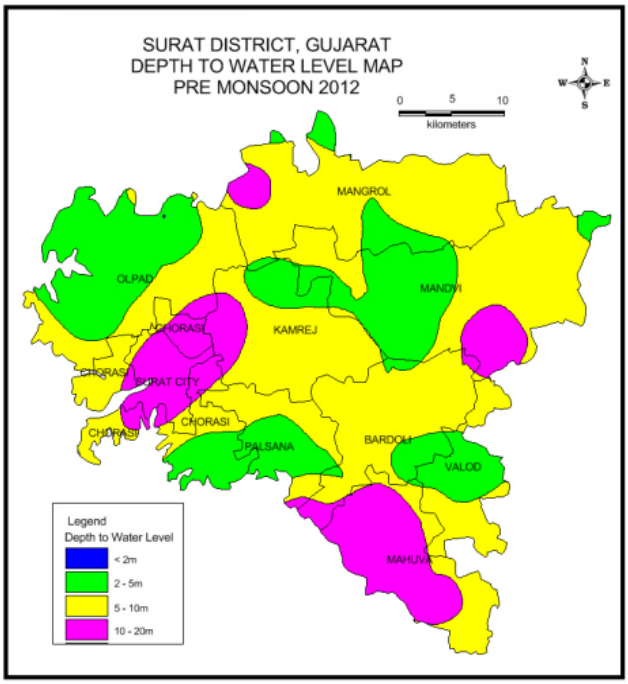

A3.2 Land Use

Surat city experienced significant urban expansion due to rapid industrialization and population growth. The supervised classification of satellite images revealed that urban areas grew by 29.5% between 1993 and 2000, 31.3% from 2000 to 2010, and 47.7% by 2019. This urbanization led to a reduction in water bodies and cultivated land, particularly during a drought in 2000, which increased barren land. The classification results, with accuracy ranging from 84% to 93.3%, highlight the environmental impact of this rapid urbanization, resulting in a shift in land use patterns and an imbalanced environmental situation in Surat.

The Tapi basin's land cover and land use characteristics highlight the region's predominant reliance on agriculture, which occupies 66.19% of the area. Key crops include wheat, sugarcane, rice, and gram, with wheat being the most significant irrigated crop, covering 30.3% of the irrigated land. Dryland farming is prevalent, with groundnut, soybean, maize, and pulses as common kharif crops, and sorghum and gram as common rabi crops.

Forests cover 24.41% of the basin, primarily consisting of dry deciduous vegetation. The culturable area constitutes 68.1% of the total area. Built-up land is minimal, at only 1.26%, while water bodies, primarily contributed by the Tapi River and its tributaries, cover 2.99% of the basin. Wastelands, including scrublands and gullied lands, account for 5.16% of the area.

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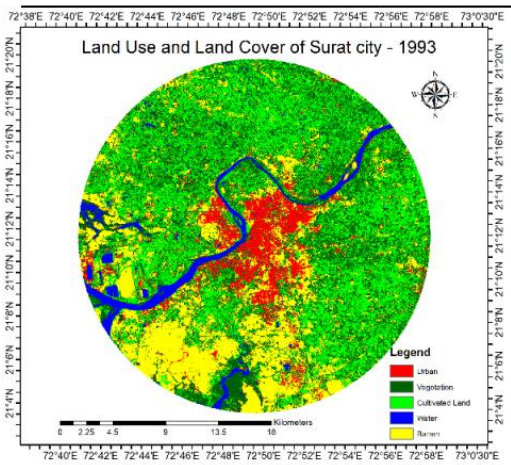
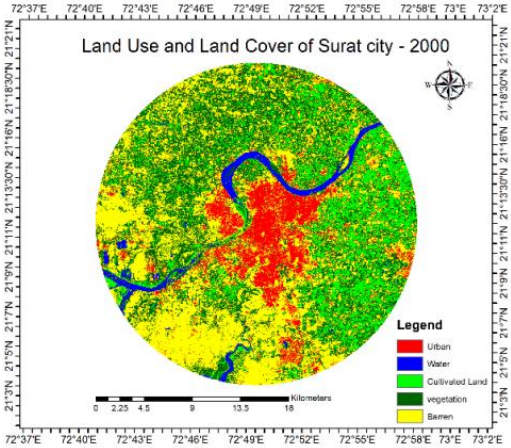
S.no.	Category	Area (Sq. km)	% of Total Area
1	Built Up Land	818.64	1.26
2	Agricultural	43117.34	66.19
3	Forest	15899.92	24.41
4	Grassland	1.47	0.00
5	Wasteland	3362.91	5.16
6	Waterbodies	1944.71	2.99

CLASS	1999 Area (km^2)	2000 Area (km^2)	2010 Area (km^2)	2019 Area (km^2)
URBAN	59.9157	77.571	112.993	164.923
WATER	29.709	21.0546	21.0357	17.6733
CULTIVATED LAND	239.953	99.7686	141.47	231.793
VEGETATION	228.562	227.209	215.748	95.3595
BARREN	148.736	281.272	215.628	197.126
TOTAL	706.875	706.875	706.875	706.875

Table: Land Use and Land Cover classification of Surat city





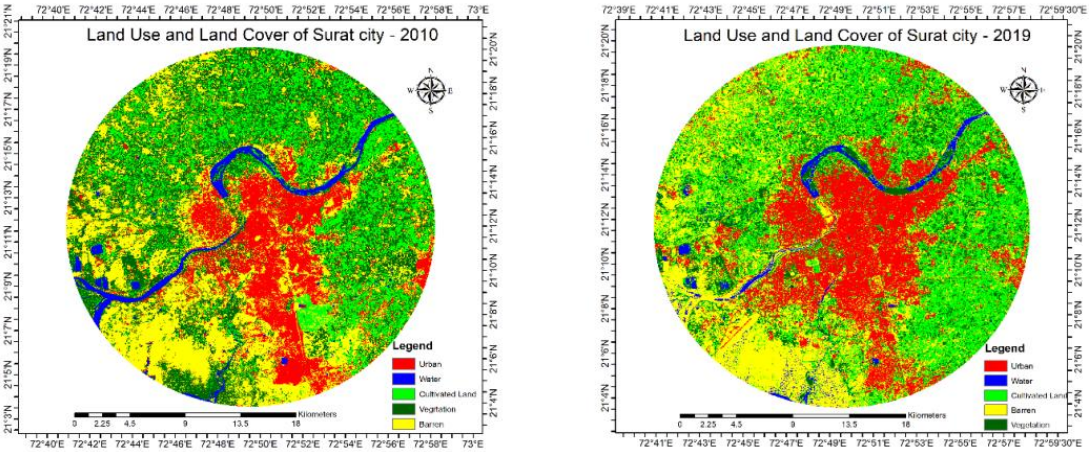


Figure: Land Use and Land Cover of Surat city

https://www.researchgate.net/figure/Land-Use-and-Land-Cover-classification-of-Surat-city_tbl3_344870260

The overlay analysis is used to understand how urban land expansion has affected water bodies in Surat. First, polygons were drawn around water bodies from a topographic map of Surat from 2011 to create a vector map showing the location and shape of these water bodies. Next, data on urban land use from 2019 was extracted from a classified satellite image of Surat. By overlaying these two datasets, the examines how much of the original water bodies from 2011 have been covered by urban development by 2019. This method helps visualize the impact of urban growth on water bodies in the city.

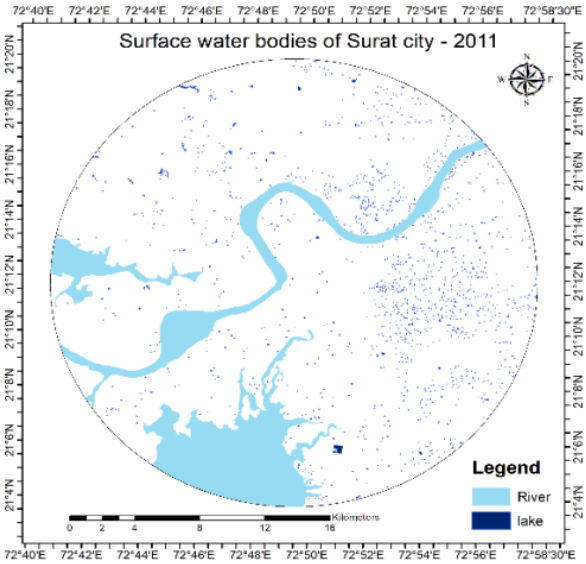


Figure: surface water bodies of Surat city

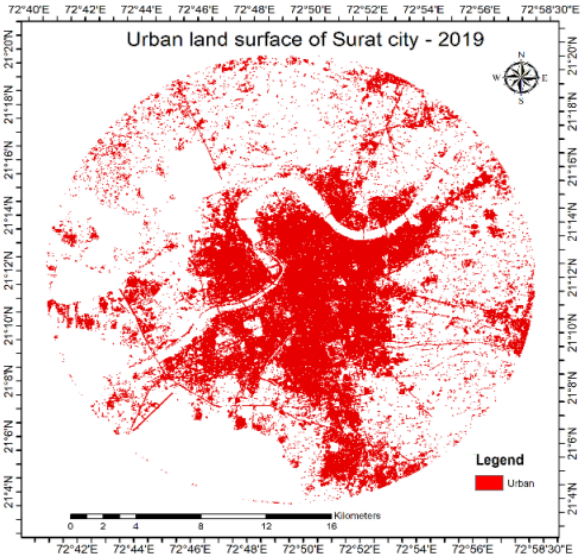


Figure: The urban land surface of Surat city

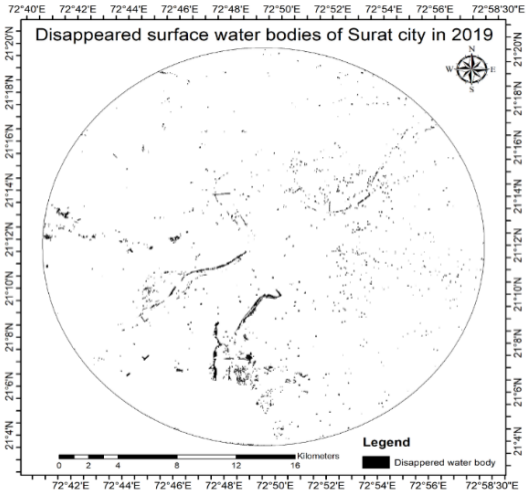


Figure: Disappeared surface water bodies of Surat city in 2019.

https://www.researchgate.net/figure/Land-Use-and-Land-Cover-classification-of-Surat-city_tbl3_344870260

Conclusion

From 1993 to 2019, Surat's urban area expanded by 175.2%, significantly altering land use, especially in the city center and fringe areas. This rapid urbanization, often at the expense of agricultural land and water bodies, has led to environmental issues such as flooding, water scarcity, and loss of natural ecosystems. About 0.812 sq.km of water bodies were converted into urban areas between 2011 and 2019. To mitigate these negative impacts, better urban planning and conservation policies are needed to protect natural resources and ensure sustainable development.

A3.3 Drainage

The Tapi is the major river which passes through the central parts of the district and flows towards the west. The river is perennial in nature. It originates in Madhya Pradesh near Betul and has about 62225 sq. Km. of catchment area. The average width of the river in the upstream of Kathor bridge is about 500m. Downstream of the bridge the average width increases to 700m. Pickup weir was constructed on the Tapi river in 1954 at Kakarapar about 56km west of Surat. Ukai dam, constructed in 1965, is situated about 25km upstream of Kakarapar weir. Other prominent rivers draining the district are Kim, Ver, Mindola, Jhankhari and Purna. The Ver flows from north-east to south-west and flows parallel to Tapi and then it flows towards west. All other rivers are situated south of Tapi and flow towards west, parallel to Tapi.

Sr. No.	Name of State	Drainage area[Sq. Km.]	Percentage of total
1	Madhya Pradesh	9,804	15.0
2	Maharashtra	51,504	79.1
3	Gujarat	3,837	5.9
		TOTAL:- 65,145	100.0

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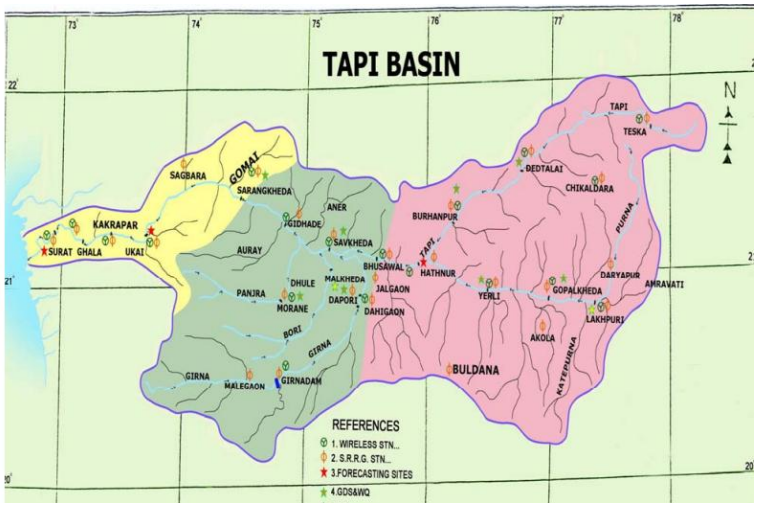
A3.4 River Basin

The Tapi is the second largest westward draining interstate river basin. It covers a large area in the State of Maharashtra besides areas in the states of Madhya Pradesh and Gujarat. The Tapi Basin is the northern - most basin of the Deccan plateau and is situated between latitudes 200 N to 220 N approximately. The Satpura range forms its northern boundary whereas the Ajanta and Satmala hills form its southern extremity. Mahadeo hills form its eastern boundary. The basin finds its outlet in the Arabian Sea in the west. Bounded on the three sides by the hill ranges, the river Tapi, along with its tributaries, more or less flows over the plains of Vidharbha, Khandesh and Gujarat. The Tapi River drains an area of 65145 Sq.Km. out of which nearly 80 percent lies in Maharashtra. The state wise distribution of the drainage area is shown in Table.

S.no.	Sub Basin	Area (Sq. Km.)	Size Range of Watershed (Sq. Km.)	No. of Watersheds
1	Tapi Lower Sub Basin	4108.90	427.18 - 781.84	7
2	Tapi Middle Sub Basin	31766.67	365.64 - 937.44	47
3	Tapi Upper Sub Basin	28047.34	322.12-937.28	46

Table: Sub-basin wise watersheds

Watershed Code	
Sub-Basin Name	**Watershed Code**
Tapi Lower	B13TAL01 to 07



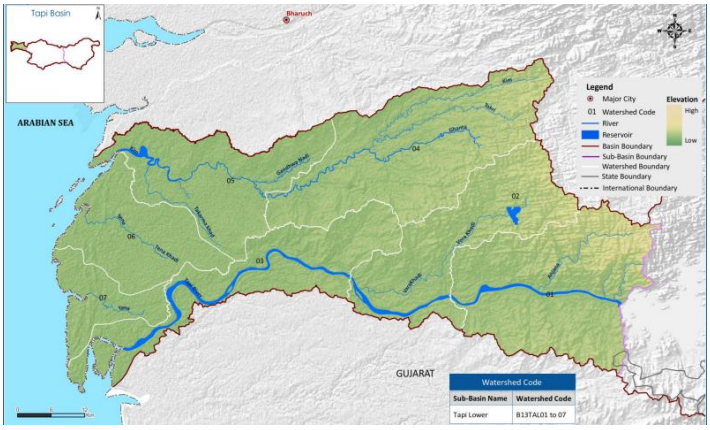


Figure: Tapi river Basin

(Source: India Water Resources Information System: www.india-wris.nrsc.gov.in.)

A3.5 Description of river system

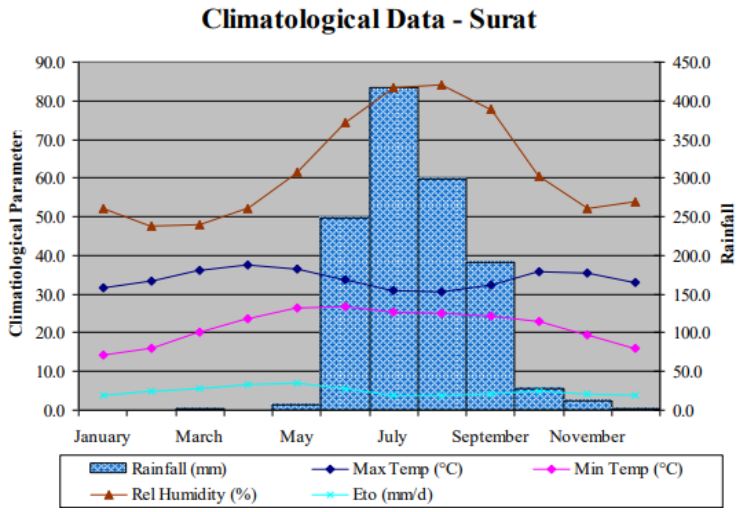
The Tapi River originates near Multai in Betul district at an elevation of 752 m above m.s.l. The total length of this west flowing river from its origin to its outflow into the sea is 724 Km. For the first 282 km the river flows in Madhya Pradesh, out of which 54 km forms the common boundary with Maharashtra State. It flows for 228 km in Maharashtra before entering Gujarat. Traversing a length of 214 km in Gujarat, the Tapi River joins Arabian Sea in the Gulf of Cambay after flowing past the Surat city. The river receives tidal influence for a length of about 25 Km upstream from the mouth. (Up to Singanpur weir causeway) The Tapi River receives several tributaries on both the banks. There are 14 major tributaries having a length more than 50 Km. On the right bank, 4 tributaries namely the Vaki, Gomai, Arunavati and Aner join the Tapi River. On the left bank, 10 important tributaries namely the Nesu, Amaravati, Buray, Panjhra, Bori, Girna, Waghur, Purna, Mona and Sipna drain into the main channel. The drainage system on the left bank of the Tapi river is, therefore, more extensive as compared to the right bank area.

A.4. Climate

(i) Type of Climate

Agro-climatic Zones: is a land unit in terms of major climate, and growing period which is climatically suitable for a certain range of crops and cultivars. The climate of the area is characterized by hot and wet summers and dry winters. The region in general has an annual rainfall of over 100cm with some local variations. Major part of the Tapi Basin falls in the Western plateau and Hilly Agro climatic zone map of Tapi basin is shown in Map10. The uppermost part of the basin falls in the “Central Plateau and Hilly region” and the lowermost part fall in the “Gujarat Plateau and Hilly region”(Source: Agro-climatic Zone: Panning Commission,1989)

https://indiawris.gov.in/downloads/Tapi%20Basin.pdf



https://www.cgwb.gov.in/old_website/District_Profile/Gujarat/Surat.pdf

A.5. Rainfall

The South-west monsoon sets in by the middle of June and withdraws by the first week of October. June to September is the period of heaviest rain. During this season, the weather is somewhat sultry and oppressive, especially in areas adjoining the Tapi River. In the post-monsoon season, a few thunderstorms occur especially in October. Therefore, the weather clears up and it is dry and pleasant throughout the Tapi valley. The Tapi basin receives its maximum rainfall in the monsoon season. The normal rainfall near the coastal areas in the basin is highest, going up to 1,000 mm. The normal annual rainfall in the Tapi basin up to is 820 mm, about 90% of which is received during the monsoon months only of which 50% is received during July and August In, the first week of October, 5.2% of the annual rainfall is received so rainfall is likely to produce a comparatively higher runoff so month of October has been included in the monsoon months. Source: (Water Year Book, 2010-2011 Central Water Commission, Ghandhinagar), Indian Meteorological Department). There are 17 districts (3-Gujarat, 4-Madhya Pradesh & 10-Maharashtra) falling in the basin which are drought prone. (Source: Drought Prone Areas Program, DPAP, MoRD, 2002).

The Upper Tapi basin had max rainfall in year 1990 is 1187.31mm while minimum annual rainfall in year 2000 is 592.4mm, The Middle Tapi basin had max annual rainfall in year of 1976 is 1427.45mm while minimum rainfall in 1991 574.94 while Lower Tapi basin had max annual rainfall of 1405mm in year of while minimum annual rainfall is recorded in year of 1991 is 578.94mm.

Rainfall Data- 2025	
District	Total Rainfall (mm)
Surat	1986.30
Tapi	1968.57

http://gsdma.org/rainfalldata-2?Type=2

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A.6. Ground Water

Groundwater quality alludes to the state of the water that is arranged under the Earth's surface. Groundwater can accumulate in breaks in subsurface rocks and the centre of soil particles. Since various blends can break down in water and others can be suspended in water, there is a potential for pollution with harmful mixes including significant particles like Electrical Conductivity (EC), Total Dissolved Solids (TDS), and many others i.e. Ca, Cl, F, Mg, NO3, SO4, and Total Hardness. This study examines Groundwater Quality in the study area of the South-West zone of Surat city, Gujarat. The South-West zone of Surat city is situated on the shoreline, causing the groundwater of the concerned area highly affected by seawater intrusion. The present study determines the Ground Water Quality Index (GWQI) from the year 2006 to 2015. It also predicts the approximate ground water quality for the next 10 years. The outcome of this study suggests the need for certain improvement in ground water quality by an appropriate method such as Managed Aquifer Recharging (MAR). It will be beneficial for the people mainly dependent on the ground water, particularly living near the coastal region.

file:///C:/Users/user/Downloads/An_assessment_of_groundwater_quality_in_South-West.pdf

(i) Description of aquifer:

The level of Ground Water Development varies from 20.14% (Olpad Taluka) to 77.72% (Choriyasi Taluka.) and overall Level of Development for the district is 40.39 %. The overall category of the district is also "Safe"

(*Refer Water Data Guide under Documents on https://www.cgwb.gov.in/cgwbpnm/public/uploads/documents/1708417527105587285 7file.pdf)*

Taluka Wise Ground Water Resources, Availability, Utilization and Stage of Ground Water Development- 2023																
DISTRICT- SURAT																
Sr No	Taluka	ANNUAL REPLENISHABLE GROUND WATER RESOURCES (Ham)				Total Annual Ground water Recharge (3+4+5+6)	Environmental Flows (ham) (5% of 7 for WTF & 10% of 7 for RIF)	Annual Extractable Ground water Resource (ham)	ANNUAL GROUND WATER DRAFT (Ham)				Allocation of Ground Water Resource for Domestic Utilisation for projected year 2025 (ham)	Net Annual Ground Water Availability for Future Use (ham)	Stage of Ground Water Extraction (%)	Categorization of Assessment Unit
		Monsoon		Non Monsoon												
		Recharge from Rainfall	Recharge from Other Sources	Recharge from Rainfall	Recharge from Other Sources				Irrigation	Industrial	Domestic	Total Draft (10+11+12)				
1	3	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1	BARDOLI	5796.8	1289.92	0	4561.71	11648.43	1164.84	10483.59	5234.10	5.35	517.66	5757.10	615.75	4628.40	54.92	safe
2	KAMREJ	9774.48	977.54	0	3365.32	14117.34	705.87	13411.47	5165.10	88.15	511.19	5764.44	608.04	7550.18	42.98	safe
3	MAHUVA	5144.86	1408.82	0	5005.87	11559.55	1155.96	10403.59	4613.80	4.46	93.72	4711.97	111.48	5673.86	45.29	safe
4	MANDVI	15216.1	1385.34	0	6741.13	23342.57	2334.25	21008.32	4751.50	41.43	583.66	5376.59	694.25	15521.14	25.59	safe
5	MANGROL	15108.39	1129.39	0	4106.60	20344.38	1017.22	19327.16	1835.40	137.80	515.75	2488.95	613.47	16740.49	12.88	safe
6	OLPAD	5405.27	1638.96	0	6896.38	13940.61	1394.06	12546.55	1235.00	54.27	0.00	1289.27	270.43	10986.85	10.28	safe
7	PALSANA	6990.26	1085.72	0	3831.06	11907.04	1190.71	10716.33	2804.10	3733.43	334.97	6872.50	398.44	3780.36	64.13	safe
8	SURAT CITY & CHORASI	4586.23	922.5	0	3665.09	9173.82	917.38	8256.44	3230.40	144.24	689.32	4063.97	819.93	4061.86	49.22	safe
9	SURAT URBAN	14073.49	0	0	0.00	14073.49	703.67	13369.82	0.00	329.94	15087.88	15417.82	17668.11	0.00	115.32	over_exploited
10	UMARPADA	13183.4	102.11	0	201.54	13487.05	674.36	12812.69	829.50	0.00	225.57	1055.07	268.31	11714.88	8.23	safe
	Surat Total	**95279.28**	**9940.30**	**0.00**	**38374.70**	**143594.28**	**11258.32**	**132335.96**	**29698.90**	**4539.06**	**18559.73**	**52797.68**	**22068.21**	**80658.02**	**39.90**	**safe**

(ii) **Unconfined Aquifers:**

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Potable: Generally, unconfined aquifers can have a mix of water quality. In Surat, potable water from unconfined aquifers can be found but may be limited in certain areas. Quality often depends on local contamination sources and management practices.

Brackish: Unconfined aquifers in Surat can contain brackish water, especially near coastal areas where seawater intrusion can impact groundwater quality.

Saline: Saline water in unconfined aquifers is also a concern, particularly in coastal zones where the intrusion of seawater can occur, leading to increased salinity in the groundwater.

Confined Aquifers:

Potable: Confined aquifers in Surat generally have better water quality compared to unconfined aquifers due to natural filtration processes through the confining layers. However, localized contamination can still affect water quality.

Brackish: The brackish water is less common in confined aquifers but can occur depending on the geological formations and proximity to saline water sources.

Saline: Saline water is less common in confined aquifers but can still be present in certain areas, particularly if there is a significant geological connection to saline water sources or if there is significant over-extraction leading to quality issues.

https://www.cgwb.gov.in/old_website/District_Profile/Gujarat/Surat.pdf

(iii) Any special quality problem, (Seawater intrusion, pollution, high fluoride etc.).

As of the latest reports from the Central Ground Water Board (CGWB), Surat District in Gujarat faces several water quality issues, including:

1. **Seawater Intrusion**: Surat, being a coastal district, is vulnerable to seawater intrusion, especially in areas where groundwater is over-exploited. This intrusion can lead to increased salinity in groundwater, affecting its potability and usability for irrigation.
2. **Fluoride Contamination**: Fluoride levels in groundwater can be a concern in some parts of Surat. Excessive fluoride can lead to health issues such as dental and skeletal fluorosis. Monitoring and managing fluoride concentrations is important to mitigate these risks.
3. **Pollution**: Industrial activities in Surat can contribute to pollution, including contamination from chemicals and heavy metals. This pollution can affect both surface and groundwater quality, making it essential to implement measures to prevent contamination and protect water sources.

file:///C:/Users/user/Downloads/An_assessment_of_groundwater_quality_in_South-West.pdf

http://jalshakti-dowr.gov.in/sites/default/files/MasterPlanForArtificialRechargeToGroundWater2020.pdf

A.7. Alternate methods

Surat and Tapi District water management methods should be tailored to local conditions such as topography, rainfall, hydrogeology, and land source water availability. Current methods are likely chosen based on these factors, with a focus on sustainability and feasibility. For high rainfall areas, rainwater

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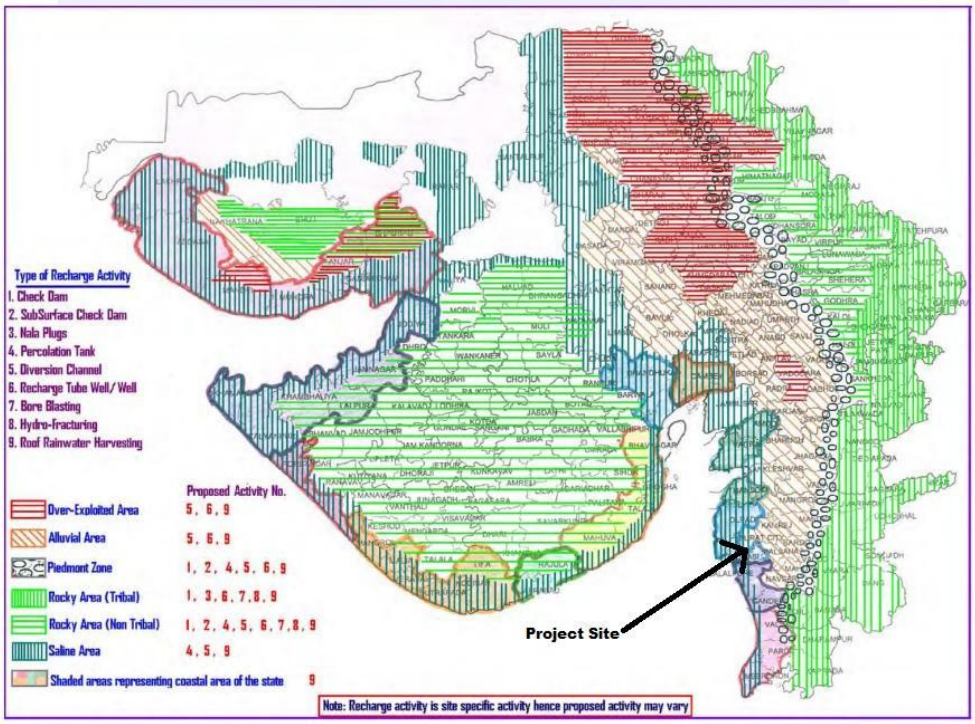
harvesting, and check dams are effective, while recharge pits and watershed management are beneficial for addressing groundwater depletion. In coastal regions facing saline intrusion, desalination and coastal aquifer management can be alternative solutions. A comprehensive approach considering these methods can enhance water resource management in Surat.

Table: Water Management Methods for Surat & Tapi District

Condition	Recommended Methods	Rationale
High Rainfall Areas	-Rainwater Harvesting - Check Dams	Captures and stores excess rainfall for dry periods.
Groundwater Depletion	-Recharge Pits - Watershed Management	Enhances groundwater recharge and manages resources holistically.
Coastal Areas with Saline Intrusion	-Desalination -Coastal Aquifer Management	Mitigates the impact of saline water on freshwater resources.

https://www.cgwb.gov.in/old_website/District_Profile/Gujarat/Surat.pdf

wasmows.gujarat.gov.in/web/public/content/cms.portal.document/164/document/Handbook_on_recharge.pdf



Number of natural and man-made water bodies including lakes in rural and urban areas in the State of Gujarat reported in the first Census of Water bodies

State	No. of water bodies including lakes 14.12.2023								
	Natural			Man made			Total		
	Rural	Urban	Total	Rural	Urban	Total	Rural	Urban	Total
Gujarat	16,166	800	16,966	36,990	113	37,103	53,156	913	54,069

sansad.in/getFile/loksabhaquestions/annex/1714/AU2017.pdf?source=pqals

A.8. Design Specifications

The **Project Proponent (PP)** initiated Decentralised Rooftop Rainwater Harvesting and Borewell Recharge Project at Dairy Units, it is being implemented across multiple dairy societies under SUMUL's operational area in Surat and Tapi districts of Gujarat.

The rainwater harvesting and artificial recharge system implemented at individual dairy units adopts **NeeRain gravity-based filtration units** integrated with existing borewells for groundwater recharge. The system is designed for decentralised operation, low maintenance, and effective pre-treatment of rooftop runoff prior to recharge.

Catchment and Conveyance System
Rainwater generated from RCC or metal sheet rooftops is collected through vertically aligned PVC down-take pipes fixed along the dairy building walls. The downpipes are hydraulically sized to handle peak monsoon runoff and convey water by gravity to the filtration unit through a horizontal PVC manifold with adequate slope to avoid stagnation and sediment deposition.

Filtration Unit Selection and Technical Specifications
Based on rooftop catchment area, one of the following NeeRain filter models is installed:

NeeRain NRN 220 (for rooftop areas up to 150 m²):

- Product Code: NRN 220
- Inlet Diameter: 110 mm
- Clean Water Outlet: 110 mm
- Drain / Impurities Outlet: 50 mm
- Minimum Water Head: 1 foot
- Principle of Operation: Reverse Y-flow, gravity-based
- Stages of Filtration: 2
- Primary Filter Element: SS 304 curved non-clogging screen
- Slot Size: 0.5 mm
- Secondary Filter Element: Polymeric screen
- Mesh Size: 200 microns
- Rate of Filtration: Up to 150 litres per minute

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- Filter Housing: ABS engineering plastic
- Filtration Efficiency: >95%

NeeRain NRU 150 (for rooftop areas up to 140 m²):

- Product Code: NRU 150
- Inlet Diameter: 110 mm
- Clean Water Outlet: 110 mm
- Drain / Impurities Outlet: 50 mm
- Minimum Water Head: 1 foot
- Principle of Operation: Reverse Y-flow, gravity-based
- Stages of Filtration: 2
- Primary Filter Element: Multilayer HDPE filter in two curved layers on PU frame
- Mesh Size: 400 microns (40 mesh)
- Secondary Filter Element: Polymeric screen
- Mesh Size: 200 microns
- Rate of Filtration: Approximately 120 litres per minute
- Filter Housing: ABS engineering plastic

First-Flush and Pre-treatment Arrangement
The NeeRain units inherently provide first-flush diversion through their reverse-flow mechanism, ensuring that initial runoff containing dust, organic matter, and rooftop debris is diverted through the impurity outlet. This prevents contamination of both the recharge structure and the aquifer.

Recharge Structure and Borewell Integration
Filtered rainwater is conveyed through a closed PVC pipeline to an existing borewell adapted for recharge. The recharge interface includes:

- A sealed inlet connection to the borewell casing
- Isolation and control valves to regulate inflow
- Structural protection around the recharge point to prevent surface contamination

The borewell functions as a vertical recharge shaft, allowing filtered rainwater to directly replenish deeper aquifer zones.

Water Quantity Estimation and Operation Period
the quantity of water diverted for recharge is estimated using:

- Effective rooftop catchment area
- Average annual rainfall of the Surat–Tapi region
- Runoff coefficients for roof material
- Filter throughput capacity and first-flush losses

Recharge occurs passively during rainfall events throughout the monsoon season, typically lasting three to four months.

Water Quality and Maintenance
the source water is rooftop runoff with low dissolved contaminants. Mechanical pre-treatment through

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the NeeRain filtration system is adequate, and no chemical treatment is required. Accessible filter housings allow routine inspection, pre-monsoon cleaning, and post-monsoon maintenance to ensure sustained system performance.

A.9. Implementation Benefits to Water Security

To achieve the groundwater recharge targets identified for the participating facilities and their surrounding areas, the project has implemented **100 individual artificial recharge structures** across dairy cooperative society locations in Surat and Tapi districts. Each structure has been installed at a separate dairy facility, allowing recharge to occur locally at the point of runoff generation.

The distribution of structure types under the project is as follows:

- **97 structures** are designed for direct recharge through existing borewells, which serve as recharge shafts.
- **2 open well recharge structures** utilize existing open wells to receive filtered rooftop runoff, allowing gradual percolation through the well walls and base to replenish shallow aquifer zones. The wells are protected against debris and surface contamination to ensure safe and effective recharge.
- **1 storage tank structure** utilizes a rainwater harvesting system with a storage tank arrangement, adopted due to site-specific suitability considerations.

The number of structures has been determined based on the availability of rooftop catchment areas at dairy facilities, the presence of suitable borewells for recharge, and the objective of capturing monsoon rainfall at source locations. Collectively, these installations are sufficient to meet the intended quantitative recharge outcomes at the facility level and contribute to groundwater replenishment across multiple talukas within the project region.

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The decentralized placement of structures across diverse locations ensures that recharge benefits are spatially distributed and aligned with local groundwater conditions, supporting sustainable aquifer management without reliance on large, centralized recharge infrastructure.

A9.1 Objectives vs Outcomes

Objective:
The objective of the project was to enhance groundwater sustainability at dairy facility locations by capturing rooftop rainwater and directing it for artificial recharge during the monsoon period. The activity aimed to reduce loss of rainwater as surface runoff, support long-term borewell performance for dairy operations, and encourage efficient water management practices across rural and semi-urban areas of Surat and Tapi districts. The project was also intended to demonstrate a practical and replicable recharge solution that could be implemented within existing infrastructure without changes to land use or natural drainage.

Outcome:
Post commencement of the activity, rainwater harvesting and artificial recharge systems were successfully implemented at multiple dairy locations, resulting in effective utilisation of monsoon rainfall for groundwater replenishment. The project has led to improved recharge of existing borewells and better management of runoff within dairy premises. Early operational experience indicates improved reliability of groundwater sources during the non-monsoon period and increased awareness among dairy members regarding water conservation. The functioning of the systems has been validated through on-site inspections, supported by photographs and schematic diagrams illustrating the flow of rainwater from rooftop collection to recharge points. These outcomes confirm that the project objectives have been met and that the installations are technically sound, functional, and suitable for long-term operation.

A9.2 Interventions by Project Owner / Proponent / Seller

To achieve the desired outcomes of improved groundwater recharge and sustainable water management at dairy facility locations, a series of targeted technical and operational interventions were carried out under the project activity. The interventions were planned to suit site-specific conditions while maintaining a uniform implementation approach across all locations.

As an initial step, suitable dairy premises were identified based on the availability of rooftop catchment areas, existing borewells or recharge points, and safe conveyance paths for rainwater. Basic site assessments were conducted to ensure that the installations would not interfere with existing operations, land use, or natural drainage patterns. Based on these assessments, standardized rainwater harvesting and recharge designs were finalized for deployment.

The core interventions implemented under the project included:

- Installation of rooftop rainwater collection systems using existing building roofs as catchments.

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- Provision of downpipes and gravity-based conveyance pipelines to safely transport rainwater from rooftops to recharge points.
- Incorporation of first-flush diversion and filtration units to remove initial contaminants and suspended solids from rooftop runoff.
- Integration of filtered rainwater flow with existing borewells or designated recharge structures to enable artificial groundwater recharge.
- Installation of inspection and control arrangements to allow routine cleaning, monitoring, and seasonal maintenance.

In parallel with physical installations, basic awareness and orientation were provided to dairy operators and staff regarding system operation, routine maintenance, and the importance of protecting recharge points from contamination. This ensured continued functionality of the systems beyond the construction phase.

Outcome post commencement:
Following implementation, the interventions resulted in effective capture and utilization of monsoon rainfall that would otherwise have been lost as surface runoff. The recharge structures began functioning during rainfall events, enabling direct contribution to groundwater replenishment at the facility level. Post-implementation observations indicate smoother runoff management within dairy premises, reduced water stagnation, and improved sustainability of borewell sources during subsequent non-monsoon periods. The combination of appropriate technical design and local-level operational awareness has ensured that the interventions are functional, durable, and capable of delivering sustained groundwater recharge benefits.

Photographic records and schematic diagrams of the installed systems are maintained as supporting evidence to demonstrate implementation and validate the outcomes achieved through these interventions.

A.10. Feasibility Evaluation

The project titled **"Borewell Recharge by The Surat-Tapi District Co-operative Milk Producers' Union Ltd., Gujarat"**, implemented by Project Proponent (**PP**), and is economically feasible due to its low capital investment and negligible recurring costs. The installations utilise existing dairy infrastructure and operate without external energy inputs, resulting in minimal operational expenditure limited to routine inspection and seasonal cleaning.

The project strengthens water reliability at dairy locations, thereby reducing potential future costs related to water scarcity or alternative water supply arrangements. Findings from comparable third-party assessments and audit reports support decentralised rainwater harvesting and recharge systems as cost-effective interventions and recommend their replication. Overall, the project delivers sustained benefits at a modest cost, making it suitable for long-term implementation and scaling.

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A.11. Ecological Aspects:

The project **"Borewell Recharge by The Surat-Tapi District Co-operative Milk Producers' Union Ltd., Gujarat"**, implemented by **PP**, contributes to multiple United Nations Sustainable Development Goals through integrated water resource management, environmental protection, and livelihood support.

Sustainable Development Goals Targeted	Most relevant SDG Target / SDG Impact	Indicator (SDG Indicator)
	Target 6.4: Increase water-use efficiency and ensure sustainable withdrawals **Target 6.6:** Protect and restore water-related ecosystems	The project captures rooftop runoff and diverts it for controlled groundwater recharge, reducing freshwater abstraction and improving aquifer sustainability. Filtration units ensure that only clean water enters the subsurface, thereby safeguarding groundwater quality.
	Target 12.2: Achieve sustainable management and efficient use of natural resources	The intervention promotes efficient use of rainwater within dairy premises, minimizing wastage and lowering dependence on groundwater extraction for operational needs.
	Target 13.1: Strengthen resilience and adaptive capacity to climate-related hazards	By storing and recharging rainwater during high-intensity rainfall events, the project reduces vulnerability to seasonal water scarcity and enhances adaptive capacity against rainfall variability and drought conditions.
	Target 15.1: Ensure conservation and sustainable use of terrestrial ecosystems	Artificial recharge supports soil moisture regimes and stabilizes groundwater-dependent ecosystems, contributing to improved land and vegetation health in surrounding areas.

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A.12. Recharge Aspects:

Please document efforts taken to ensure that the *Quality of Surplus Recharge water is clean, free from contamination and has compatibility with quality of native ground water in aquifers.*

A.12.1 Solving for Recharge

Ultimately, the volume of groundwater recharge benefit to the subbasin is the most critical aspect for such MAR activities. Groundwater recharge is quantified as the deep percolation of surface water applied during project implementation. Using a field-scale water budget, deep percolation can be calculated as the difference between all other inflows and outflows, per the equation below, with each other inflow and outflow being quantified:

Recharge = Rainfall + Surface Inflow − Evapotranspiration − Surface Outflow − Change in Storage

Evapotranspiration & Other Data: https://datameet-pune.github.io/open-water-data/docs/open-water-data-paper.pdf (or available under Documents Section- Water Data Guide)

Root Zone = The root zone is comprised of the upper portion of the soil where water extraction by roots occurs, above the depth at which water infiltrates to the groundwater system. The depth to the bottom of the root zone varies by crop, but typically extends up to seven feet.

Surface Inflow= Surface inflows can be either directly measured or calculated from measured values. In fields directly served by metered lift pumps or metered gates, the volume of surface inflows to the field can be directly measured or calculated from totalized measurements. Typical accuracies of pipe flow measurements range from 1-12 percent. In fields that are indirectly supplied with surface water, surface inflows may need to be calculated from upstream and downstream flow measurements, or through theoretical or empirical equations relating available data to field surface inflows. For example, fields served from canals measured using weirs, or fields served from canals that deliver water to multiple locations downstream of a measurement device may require site-specific calculations to quantify surface inflows to a specific field. Low-cost in-field measurements can also be made by setting up flashboards at the measurement location and correlating the "runup" of an unsubmerged weir overflow on a flat weir stick to the flow rate using standardized equations. Typical accuracies of "runup" or indirect flow measurements may exceed 10 percent, depending on site conditions and the accuracy of measurement data.

To monitor surface inflows, project owners may record flow data, maintain irrigation logs, and maintain logs of any other parameters required to calculate field deliveries, depending on the unique conditions of their field. Project owners may also consider using mobile flow monitoring equipment to measure or verify surface inflows.

Surface Outflows: To monitor surface outflows, users may record flow data or water level data and maintain logs of any other parameters required to calculate outflows, depending on the unique conditions

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of their project activity. Pressure transducers and dataloggers may be used to automatically monitor water levels, or users may install wooden stakes to manually monitor water depths.

Change in Storage = The change in surface storage, or average ponded water depth, can be calculated from measured and observed changes in water surface levels at points throughout the project field. Over the annual project implementation period, **the total change in surface storage is typically zero**, provided that the surface of a field is dry and free of ponded water at the start and end of the project.

While the uncertainty of each inflow and outflow will vary based on field conditions and measurement devices, typical uncertainties associated with each water budget component are summarized in the table below. The uncertainty of deep percolation (i.e., recharge) can then be calculated from these other uncertainties, for example following the process described by Clemmens and Burt (1997). Users can use the following table to eliminate uncertainty from their estimates.

Water Budget Component	Typical Estimated Uncertainty (%)	Description
Surface Inflow	NA (0%)	No canal, river, or external surface water diversion involved
Precipitation	5%	Rainfall estimated using IMD / district rain gauge data for rooftop catchments
Surface Outflow	10%	Minor overflow possible during extreme rainfall events
Evapotranspiration	Negligible (<5%) / Not Considered	No open storage; immediate conveyance to recharge
Change in Storage	Negligible (<5%) / Not Considered	Subsurface storage change inferred from recharge into aquifer
Deep Percolation	5%	Recharge through borewells/open wells into variable subsurface strata
Total		**20%**

Other factors of uncertainty to consider when quantifying recharge are:

- Deep percolation does not immediately recharge the groundwater system. There is a time lag between when deep percolation occurs through the root zone and when that water reaches the saturated groundwater system.
- Subsurface inflows and outflows can occur through the groundwater system. While deep percolation may supply water to the groundwater system, that water may migrate away from the field along groundwater gradients.

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Groundwater recharge can also be monitored and verified through groundwater level measurements at groundwater wells adjacent or near to the project activity. For instance, groundwater level measurements collected before, during, and after implementation can potentially help verify that net recharge is occurring; especially in well-positioned wells with continuous monitoring.

A.13. Quantification Tools

The baseline scenario is the situation where, in the absence of the project activity, unutilized rainwater flows uncollected into drains or is not conserved and harvested within the project boundary and hence remains unutilized. Baseline scenario, if not directly measurable, is calculated by using the UWR Standard. The PP has selected the following method from UWR standard

Harvesting potential or Volume of water utilized (m3) = Area of Catchment/Roof/Collection Zone (m2) X Amount of rainfall (m) X Runoff coefficient

As per UWR Standard

Surface area	Runoff Coefficient (K)
Industrial (Asphaltic or concrete pavement)	0.9

Area of Catchment:

Annual Rainwater harvesting Potential

Annual rainwater harvesting potential is given by $V = K \times I \times A$

- Where, V=Volume of water that can be harvested annually in m^3.
- *K = Runoff coefficient*
- *I = Annual rainfall in (m)*
- *A = Catchment area in (m^2)*

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Quantification of RoUs

Year	Crediting Period	RoUs (1000 Liters)/Year
2025	26/05/2025-31/12/2025	25953.34678
Total		**25953.35**

Sr. no.	DCS_Code	DCS_Name	m2	Rainfall	Run off Coefficient	Uncertainty Factor	RoU
1	473	Ukhalda	285.7088	1968.57	0.9	0.8	405
2	464	Kelai	194.3643	1968.57	0.9	0.8	275
3	188	Kasal	123.6653	1986.3	0.9	0.8	177
4	182	Uteva	136.701	1986.3	0.9	0.8	196
5	115	Vakilpura	104.5464	1986.3	0.9	0.8	150
6	171	Vahar	69.5827	1986.3	0.9	0.8	100
7	687	Chavada	50.752	1986.3	0.9	0.8	73
8	608	Juna Chakara	42.2532	1986.3	0.9	0.8	60
9	794	Shaktipura	217.462	1986.3	0.9	0.8	311
10	664	Dharmpor	154	1986.3	0.9	0.8	220
11	952	Khambhala	55.815	1968.57	0.9	0.8	79
12	887	Vadi-Faliyu-Dhamodala	83.7225	1968.57	0.9	0.8	119
13	1028	Jagruti-Mahila	154.696	1986.3	0.9	0.8	221
14	254	Nihali	195.1543	1986.3	0.9	0.8	279
15	257	Shri Gopala	185.928	1986.3	0.9	0.8	266
16	825	Dhundhesa	223.138	1986.3	0.9	0.8	319
17	291	Kachal	139.446	1986.3	0.9	0.8	199
18	450	Jamankuva	111.63	1968.57	0.9	0.8	158
19	914	Khokhasa	106.0485	1968.57	0.9	0.8	150
20	446	Sandhkuva	156.16	1968.57	0.9	0.8	221
21	303	Mahuvaria	1115.6291	1986.3	0.9	0.8	1596
22	3	Tadad	486.7222	1986.3	0.9	0.8	696
23	682	Taparvad	123.6653	1968.57	0.9	0.8	175
24	820	Angaldhara	306	1986.3	0.9	0.8	438
25	423	Mota Tarpada	88.3158	1968.57	0.9	0.8	125
26	951	Temka	119.072	1968.57	0.9	0.8	169
27	772	Virathava	67.045	1968.57	0.9	0.8	95
28	770	Malangdev	144.9604	1968.57	0.9	0.8	205
29	255	Dhorikui	178.9359	1986.3	0.9	0.8	256
30	493	Tapi-Khadakala	100.406	1968.57	0.9	0.8	142
31	714	Kapadbandh	124.928	1968.57	0.9	0.8	177
32	603	Vadi-Rupgadh	64.1415	1968.57	0.9	0.8	91
33	1166	SrijiDairy Farm	240.8098	1986.3	0.9	0.8	344
34	136	Nasarapur	156.8	1986.3	0.9	0.8	224
35	804	Kosambia	146.3924	1968.57	0.9	0.8	207
36	573	Tarbhon	320	1986.3	0.9	0.8	458

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37	573	Tarbhon-Member	480	1986.3	0.9	0.8	686
38	692	Amalpada	97.6305	1968.57	0.9	0.8	138
39	720	Mandavipani	97.6305	1968.57	0.9	0.8	138
40	62	Motibhatlav	232.2576	1986.3	0.9	0.8	332
41	694	Satkashi	159.942	1968.57	0.9	0.8	227
42	721	Juni Bavali	119.072	1968.57	0.9	0.8	169
43	161	Unchavan	125.538	1968.57	0.9	0.8	178
44	161	Unchavan-Member	185.928	1968.57	0.9	0.8	264
45	1120	Kacholi	195.261	1968.57	0.9	0.8	277
46	651	Akalutara	74.42	1968.57	0.9	0.8	105
47	618	Navibhilbhavali	97.6305	1968.57	0.9	0.8	138
48	659	Junai	69.723	1968.57	0.9	0.8	99
49	1050	Samarkuva	97.9965	1968.57	0.9	0.8	139
50	923	Amaldi	60.4266	1968.57	0.9	0.8	86
51	676	Bhattwada	106.0485	1968.57	0.9	0.8	150
52	701	Shri Sarjamli	130.174	1968.57	0.9	0.8	185
53	627	Limbi	176.6316	1968.57	0.9	0.8	250
54	272	Kos	725.2138	1986.3	0.9	0.8	1037
55	518	Chitalada	63.196	1986.3	0.9	0.8	90
56	969	Sevati	71.4432	1968.57	0.9	0.8	101
57	414	Dhajamba	323	1968.57	0.9	0.8	458
58	928	Bhanpur	270	1968.57	0.9	0.8	383
59	328	Kalamkui	148.84	1968.57	0.9	0.8	211
60	333	Khusalpura	130.8816	1968.57	0.9	0.8	186
61	457	Nishana	185.928	1968.57	0.9	0.8	264
62	471	Dumda	209.625	1968.57	0.9	0.8	297
63	458	Chorvad	144.7919	1968.57	0.9	0.8	205
64	620	Kavala	113.8489	1968.57	0.9	0.8	161
65	444	Dhamodi	209.6479	1968.57	0.9	0.8	297
66	525	Nani-Khervan	129.654	1968.57	0.9	0.8	184
67	544	Moti-khervan	213.6648	1968.57	0.9	0.8	303
68	700	Panchpipla	255	1968.57	0.9	0.8	361
69	329	Dhamodala	162.6108	1968.57	0.9	0.8	230
70	468	Kanara	164.0412	1968.57	0.9	0.8	233
71	465	Amlipada	159.4449	1968.57	0.9	0.8	226
72	325	Bedkuva	119.6134	1968.57	0.9	0.8	170
73	376	Katiskuva_Najik	131.8866	1968.57	0.9	0.8	187
74	401	Karnjvel	457.8363	1968.57	0.9	0.8	649
75	678	Jamapur	272	1968.57	0.9	0.8	386
76	763	Bedkuva-Najik	331.837	1968.57	0.9	0.8	470
77	1051	Kelvan	120.5604	1968.57	0.9	0.8	171
78	822	Kakadava	160	1968.57	0.9	0.8	227
79	436	Tokarava	169.6365	1968.57	0.9	0.8	240
80	426	Gopalpura	460.1444	1968.57	0.9	0.8	652
81	34	Ishanpor	139.446	1986.3	0.9	0.8	199

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82	316	Nalotha	162	1968.57	0.9	0.8	230
83	177	Khodamba	144.1804	1986.3	0.9	0.8	206
84	183	Vadesia	167.384	1986.3	0.9	0.8	239
85	250	Moticher	297.558	1986.3	0.9	0.8	426
86	566	Karchelia	130.174	1986.3	0.9	0.8	186
87	631	Kavitha	111.63	1986.3	0.9	0.8	160
88	565	Gavan	83.7225	1986.3	0.9	0.8	120
89	897	Kalijaman	46.482	1986.3	0.9	0.8	66
90	567	Servan-Fokadi	55.7784	1986.3	0.9	0.8	80
91	25	Amboli	143.0847	1986.3	0.9	0.8	205
92	168	Sevalan	41.8155	1986.3	0.9	0.8	60
93	584	Amkuti	83.7225	1986.3	0.9	0.8	120
94	901	Gaybeeumar	196.3976	1968.57	0.9	0.8	278
95	475	Devara	181.396644	1968.57	0.9	0.8	257
96	894	Ranikhdakala	288.952644	1968.57	0.9	0.8	410
97	151	Navachakra	374	1986.3	0.9	0.8	535
98	811	Chikhali-Bhensrot	255	1968.57	0.9	0.8	361
99	110	Mangrol	223.138	1986.3	0.9	0.8	319
100	134	Ognisa	125.3094	1986.3	0.9	0.8	179
TOTAL							**25953**

A.14. UWR Rainwater Offset Do No Net Harm Principles

Under the **Rooftop Rainwater Harvesting and Borewell Recharge Project implemented by PP** runoff generated from dairy rooftops and paved premises is captured and diverted for groundwater recharge instead of being discharged into surface drains. The collected runoff is routed through basic filtration units and conveyed into existing borewells to augment aquifer recharge. This approach supports groundwater availability during non-monsoon periods. Members associated with village dairy societies are involved in routine supervision, system upkeep, and record-keeping, strengthening their participation in local water resource management.

A.15. Scaling Projects-Lessons Learned-Restarting Projects

Recent experiences with water supply interruptions and conservation challenges in **Surat District** highlight ongoing pressures on local water resources. In late 2023, residents across different zones of Surat city faced extended periods of low tap water supply, requiring temporary reliance on tankers and highlighting limits in conventional water distribution systems. Additionally, Surat's water authorities have taken steps to improve conservation outcomes citywide, earning recognition under national water stewardship initiatives due to integrated water management efforts. These developments underscore that even well-managed urban systems can experience stress, especially during dry spells or infrastructure maintenance.

Against this backdrop, the **Rooftop Rainwater Harvesting and Borewell Recharge Project implemented by The Surat-Tapi District Co- operative Milk Producers' Union Ltd. (SUMUL)** can be scaled further in a way that complements both rural groundwater needs and broader urban water management.

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The project's expansion can build on existing practices and avoid duplication by:

- **Extending implementation to additional dairy cooperative facilities** where rooftop runoff exists and recharge opportunities are technically feasible, enabling larger cumulative recharge across talukas without infrastructure redesign.

- **Integrating with government water conservation programs** (such as district groundwater recharge campaigns and municipal rainwater harvesting initiatives) to ensure that efforts in individual facilities add value to area-wide water balance objectives rather than operating in isolation.

- **Collaborating with urban water supply and stormwater management systems** to capture untapped runoff from larger roofed surfaces and paved dairy precincts before it enters storm drains, aligning recharge objectives with urban flood mitigation and stormwater management.

- **Sharing standardised implementation guidelines and monitoring protocols** developed under the project with local authorities and other cooperative sectors to harmonise technical standards and reduce repeated trial-and-error installations.

By aligning with existing integrated water resource practices, engaging with public conservation efforts, and optimizing recharge opportunities at both facility and landscape scale, the project can enhance water sustainability outcomes for both rural dairy operations and surrounding urban communities.

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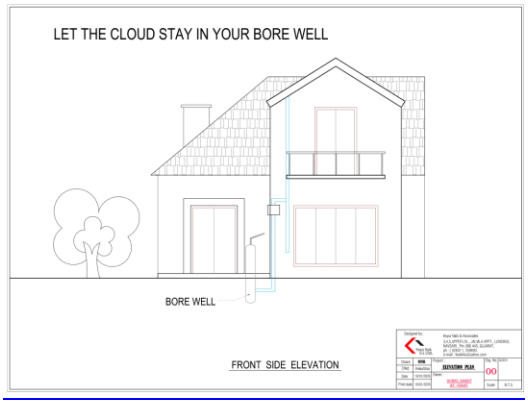
Appendix: Layout & Relevant Images



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Appendix: Completion Certificate

રજી.નં.સે. ક્રુ. ૧૫૦૮૯

શ્રી ઉખલદા દૂધ ઉત્પાદક સહકારી મંડળી લી.

કોડ નં. ૪૭૩

મુ. ઉખલદા, તા. સોનગઢ, જી. તાપી.

Mo. 7567515586

Mo. 8141442140

મંત્રીશ્રી

Date : 19.02.2026

તારીખ : 25/12/26

To Whomsoever It May Concern

Subject: Certificate of Completion – Rooftop Rainwater Harvesting System

Sir/Madam,

This is to inform you that at our Ukhalda Village Dairy Cooperative Society, the Rooftop Rainwater Harvesting System has been successfully installed under the Water Conservation Project supported by SUMUL Dairy. The installation work was completed on __/05/2025.

The system is fully operational and available for regular use of the society. Under this project, the necessary pipeline, filter fittings, and related installation work for collecting rooftop rainwater and recharging it into the borewell have been properly completed.

The total expenditure incurred for this installation is ₹ 23535

Key Benefits Observed from Installation:

- Groundwater level improvement due to borewell recharge.
- Improvement in borewell water quality through filtration of rainwater.
- Positive environmental impact through water conservation.
- Increased awareness about water conservation in the village.

Additional Details:

1. Society Details

- Village: Ukhalda
- Taluka: Songadh
- District: Tapi
- Latitude: 21.9214815
- Longitude: 73.4882673

2. Rooftop Area Details for Rainwater Collection

- Roof Length (L): 17 meter
- Roof Breadth (B): 19 meter

Yours faithfully,

ઉખલદા દૂધ ઉ. સ. મં. લિ.

મંત્રી

॥ વિના સહકાર નહિં ઉદ્ધાર ॥

ધી વહાર દૂધ ઉત્પાદક સહકારી મંડળી લી.

મુ. પો. વહાર, તા. ઉમરપાડા, જિ. સુરત. પીન નં. ૩૯૪૪૪૦

રજી.નં. ૧૬૪૮૯, રજી તા. ૧૨-૨-૮૮

જા.નં.

To whomsoever it may concern તારીખ 02-2026

Subject: Certificate of completion - Rooftop Rainwater Harvesting System

Sir/Madam

This is to inform you that at our Vahar Village Dairy Cooperative Society the Rooftop Rainwater Harvesting System has been successfully installed under the water conservation project supported by Sumul Dairy. The installation work was completed on 21/05/2025

The system is fully operational and available for regular use of the society. Under this project the necessary pipeline filter fittings and related installation work for collecting rooftop rainwater and recharging in into the borewell have been properly completed.

The total expenditure incurred for this installation is ₹ 11865

Key Benefits observed from installation:

- Groundwater level improvement due to borewell recharge
- Improvement in borewell water quality through filtration of rainwater
- Positive environmental impact through water conservation.
- Increased awareness about water conservation in the village.

Additional Details

1. Society Details
- Village - Vahar
- Taluka - Umarpada
- District - Surat
- Latitude - 21.4263952 00
- Longitude - 73.3573035 00

(2) Rooftop area Details for Rainwater collection

Roof Length (L) — 10.37

Roof Breadth (B) — 6.71.

ધી વહાર દૂધ ઉત્પાદક સ. મં. લી.

મંત્રી